

Managerial Accounting Final Project:
Aztec Airwalk, Tre Flipping Through Your Skate Spot

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Executive Summary

Aztec Airwalk is a collective of SDSU alumni who design and manufacture skateboard ramps for a national audience, with a focus on the Southern California skateboarding community. We want to expand the San Diego skating scene, develop local talent, and highlight local skate features by serving lower-income households and skaters without access to existing local infrastructure.

We offer bespoke solutions but primarily have a core suite of ramps, which include launch ramps, bank ramps, quarterpipes, and funboxes. We also offer premium upgrades for those ready for the next level, including more robust rails and coping, and smoother, more durable skate surfaces. Free shipping is offered nationwide; however, the true transportation costs are absorbed and distributed by the unit price of each ramp. Our initial recommended product mix was 25% funboxes, 20% bank ramps, and quarterpipes, respectively, 10% launch ramps, then 15% skatelite upgrades, and 10% coping upgrades. Upon CVP analysis, Funboxes (37.2%) and Skatelite upgrades (43.9%) will be our key revenue and profit drivers.

Company Description

We are Aztec Airwalk, a San Diego-based producer of skateboarding infrastructure. A collaborative effort among SDSU skateboarding enthusiasts, we were raised under Birdhouse, Sector 9, and Plan B Skateboards. Our vision is to make San Diego a powerhouse destination for street skating in the community and to develop local talent to reach their full potential. We strive to achieve this through building low-cost, approachable quality skateboard ramps that anyone can afford. We lower the barriers to entry for lower-income families, get kids away from iPads and TVs, and out into the street. We don't want to cage and consolidate them in small, neglected

parks. We want them to turn the street into a playground, see their surroundings from new perspectives.

The ramp types to support this mission are (a) launch ramps, (b) bank ramps, (c) quarterpipes, and (d) funboxes. This product mix ensures that our customers have a wide spectrum of options to augment their environment for skill building and exploration. For those who want a lux experience, premium upgrades are offered. This includes improved skate surface material (via Skatelite) and higher-quality, weather-resistant coping to give your skaters years of problem-free grinds. Your squad can handplant on aging infrastructure without fear of contracting tetanus.

Our current target market is San Diego's lower-income households with children. There are parks scattered throughout San Diego, from Carlsbad to Barrio Logan. More concrete bowls and pools than anyone can skate in a lifetime. But there's plenty of opportunity in nearby neighborhoods, in schools, parks, and business centers. San Diego deserves its own El Toro stairs. We use a focused, differentiation competitive advantage strategy, aiming to become a household name in San Diego through community engagement, talent development, and bespoke infrastructure options for different street features. Though our core catalog will serve as the primary generator of revenue.

Product Costing and Bill of Materials (BOM)

When researching potential ramp plans, we first considered multiple free, online resources: DIY Skate, Xtreme Skater, and Instructables. These were not sufficient for our desired depth. Upon deeper research, we found a definitive guidebook on how to build DIY skate ramps from one of the leading skateboard media authorities, Thrasher. A copy of *Thrasher presents: How to Build a Skateboard Ramp* was checked out from North Clairemont Library, which

provided thorough breakdowns of how to build various skateboard ramps. The raw material costs were a combination of easily available hardware stores: Home Depot and Lowe's. They had a variety of similarly specified materials, so we selected based on the material requirements and the lowest cost.

Labor was similarly multifaceted. For man-hour requirements, though we had step-by-step instructions for each ramp, it was difficult to eyeball approximately how long each task would take. We turned to historical data and expert judgement, referring to those same free, online DIY ramp sites that had analogous ramp designs and labor estimates. The wage rates were a combination of various sources, such as Glassdoor and Indeed, centered specifically around San Diego's labor market.

Pricing Strategy

Cost-Plus is the primary pricing strategy we'll employ at launch, with a flat 40% markup on the total unit cost (see Workbook). "Downside" and "Upside" scenarios and Target Contribution Margin % (set at 30%) are also used to influence the pricing method. A/B Testing will be applied in phases, releasing variant mixes and iteratively adjusting our baseline unit prices per actual demand insight. These values were inspired by parsing various financial statements in adjacent industries (outdoor recreational equipment manufacturers, furniture builders, sports equipment makers) and deriving their typical contribution margins. This flat cost-plus pricing is applied to the accessories as well; the coping upgrade is similarly sourced and priced from local hardware stores. However, the Skatelite surface material requires heavy waste-to-energy (WE) processing, is sourced from the proprietary Skatelite vendor, and has high total unit variable costs for the manufacturer.

Shipping Costs and Regional Strategy

Overview

This section creates the shipping model, the zoning structure for the various regions, allocates freight costs, and performs the cost-volume-profit (CVP) analysis for our company's ready-to-assemble skateboard ramps. We have two starting assumptions: that all our shipments will originate from a single warehouse in the San Diego metropolitan area and that all customers receive "free shipping" – meaning that the transportation costs are embedded in the product pricing rather than being charged separately by region.

Integration of BOM into Shipping Assumptions

Our BOM for the company's primary ramps was used to estimate the realistic weights for each ramp. Industry-standard material densities of 3/4" and 3/8" plywood sheets, 1/4" Masonite or Skatelite riding surface, 8-foot 2x4 framing lumber, steel or stainless coping, nails, screws, and other hardware were used to convert the totals of the various designs into weights. This analysis resulted in a reasonable estimate for shipping weight for each unit, which would determine the final freight cost assumptions.

We will use the quarter pipe as an example. The BoM shows that the quarter pipe uses multiple sheets of 3/4" and 3/8" plywood, a Skatelite riding surface, several 2x4x8' framing members, and steel coping with hardware. Industry references indicate that a 4x8' sheet of 3/4" plywood weighs approximately 60-70 lbs and 3/8" plywood weighs roughly 36-44 lbs per sheet. Skatelite pro 4x8' panels are 1/4" thick and similar in weight to a dense engineered wood sheet at about 50 lbs per sheet. A standard 8-foot 2x4 board weighs around 8-9 lbs, depending on moisture content. Multiplying these weights by the quantity of each component and adding 20-

30 lbs for steel coping and fasteners results in an estimated quarter pipe shipping weight in the 400–450 lbs range. For planning purposes, we use an approximate estimate of 425 lbs.

A similar analysis can be used for the other materials in the BoM, and results in an estimated 337 pounds for the bank ramp, 2,475 pounds for the much larger funbox module (most likely shipped on multiple pallets), and about 250 pounds for the launch ramp.

Industry-standard plywood weights range from 60–70 lb for 3/4" panels and 36–44 lb for 3/8" panels (Chanta Plywood, n.d.; Soundproofing Company, n.d.; Beatty Lumber Company, n.d.). Skatelite Pro 4×8 sheets weigh approximately 50 lb (Skatelite, n.d.). A kiln-dried 2×4×8 ft board weighs approximately 8–9 lb depending on moisture content (Inch Calculator, n.d.).

Crate Dimensions and Freight Class Considerations

The assembled ramp dimensions from the BoM were translated into flat-pack crate dimensions. This assumes that all the panels are cut from standard 4x8 sheets and then stacked together. For the quarter pipe, this results in an estimated crate footprint of 96x48" with a stack height of about 20". The bank ramp is modeled as a 72x48x18" crate, the launch ramp as a 72x48x16" crate, and the funbox as two pallets at 96x48x30" due to the large number of sheets and framing members.

These proposed dimensions allow the ramps to fall into typical less than truckload (LTL) freight classes based on the densities and the handling characteristics. Dense, crated wood and steel structures generally fall in the Class 70 – 100 range, which is consistent with how LTL carriers price crated machinery or wooden fixtures. The shipping model assumes LTL shipment of each crate rather than full truckload moves, which is appropriate for the projected volume.

Typical LTL freight classes for dense crated wood and steel structures fall within Class 70–100 according to NMFTA density-based standards and common carrier guides (NMFTA, n.d.; FreightCenter, n.d.).

Zone Structure from a San Diego Origin

The United States was broken into four zones, as described by the project brief:

1. West of the Rockies
2. South
3. Midwest
4. Northeast

These zones align with increasing transit distance and freight cost. The West of Rockies zone includes California and neighboring western states and represents the shortest and least costly shipping lanes. The South and Midwest are moderate to long haul LTL lanes, while the Northeast represents the longest transit distances and the highest base rates.

Freight Cost Estimates and 'Free Shipping' Allocation

The costs for freight were estimated by referencing typical LTL market ranges for palletized freight. Current industry guides indicate that LTL freight often falls in the \$0.20-0.45 per pound range, with typical pallet-level costs between approximately \$120 and \$250 per shipment, depending on distance, weight, and service level. Using those ranges as a benchmark and layering on conservative assumptions for residential delivery and liftgate service, the following per crate costs were adopted for a representative ramp shipment originating in San Diego: \$55 to the zone West of the Rockies, \$100 to the South, \$115 to the Midwest, and \$140 to the Northeast.

We cannot price discriminate by location; our model allocates shipping costs via a weighted average method. Assuming a sales mix of 40% West, 25% South, 20% Midwest, and 15% Northeast, the weighted average shipping cost per unit is approximately \$91. This value is

treated as part of the unit variable cost in the CVP model and must be recovered through the contribution margin.

Facilities, Equipment, and Fixed Costs

The CVP model also incorporates fixed costs related to the manufacturing facility and equipment. Market reports for the San Diego region show average industrial asking rents of about \$1.60 – 1.70 per square foot per month on a triple-net basis. A conservative placeholder of \$225,000 approx per year was used in the CVP analysis as a fixed cost placeholder.

San Diego industrial asking rents average \$1.50 per square foot per month (CBRE, 2024). A 2,000 sq ft warehouse at market rates therefore costs roughly \$36,000 annually in rent alone, and when utilities, insurance, equipment depreciation, and administrative overhead are included, a conservative fixed-cost placeholder of \$225,000 per year is justified.

Employee and Labor Cost Analysis

A well-designed labor structure is essential for Aztec Airwalk's manufacturing efficiency, cost accuracy, and internal control environment. The organization requires a balance of direct labor, those who physically build the ramps, and indirect labor, those who support operations through administrative, managerial, and procurement activities. Labor classifications (direct vs. indirect; fixed vs. variable) were determined using managerial accounting principles and cross-referenced with wage benchmarks from the U.S. Bureau of Labor Statistics to ensure realistic modeling of a San Diego-based manufacturing operation.

Organizational Roles and Labor Classification

Aztec Airwalk employs six workers, each serving a specific operational purpose:

- General Manager (Indirect, Fixed): Provides overall operational oversight, supervises administrative and procurement staff, ensures production goals are met, and maintains

safety and financial controls. Because this is a salaried leadership role whose workload does not scale with ramp output, it is classified as fixed indirect labor.

- **Production Lead (Direct, Variable):** Oversees the daily workflow of ramp production, trains production specialists, ensures accurate cutting, drilling, and assembly, and performs quality control checks. This role directly contributes to ramp construction and varies with production volume, classifying it as direct variable labor.
- **Production Specialists (2 roles) – Direct, Variable:** These workers perform the core physical tasks of ramp manufacturing, including cutting plywood and lumber, drilling pilot holes, assembling structural frames, attaching surfaces, and packing finished products. Their hours increase as production increases, making them direct variable labor.
- **Procurement & Inventory Coordinator (Indirect, Variable):** Manages purchase orders, incoming shipments, inventory levels, and material staging for production. Although not directly building ramps, this role’s workload increases as materials are consumed, aligning with indirect variable labor.
- **Accounting & Administrative Manager (Indirect, Fixed):** Handles payroll, bookkeeping, vendor payments, customer billing, regulatory paperwork, and customer communication. Since administrative tasks remain relatively consistent regardless of production volume, this role is indirect fixed labor.

Labor Cost Determination

Hourly wages were benchmarked to the San Diego labor market using BLS occupational wage data. Each role was assigned 2,080 annual hours (40 hours per week, 52 weeks per year). Direct labor cost per unit is determined by multiplying labor hours per ramp by the applicable hourly wage rates for production employees.

Labor Hours per Unit

Labor hours per unit were determined by decomposing the production workflow into cutting, assembly, and packing activities. Cutting time was estimated based on the number of plywood sheets and 2x4 lumber pieces per BOM; assembly time reflects drilling, coping installation, framing, and attaching surfaces; and packing time includes crating, labeling, and banding products for shipment.

Final estimates:

- Quarter Pipe: 2 hours
- Bank Ramp: 3.0 hours
- Launch Ramp: 4 hours
- Funbox: 6.0 hours
- Accessory 1 (Skatelite Upgrade): 0.5 hours
- Accessory 2 (Coping Upgrade): 0.5 hours

These hours were applied directly in the ABC sheet to determine total labor hours consumed and facility overhead absorbed per product.

Indirect Labor and Overhead Allocation

Indirect labor costs (General Manager, Procurement, and Administrative staff) were excluded from direct labor and instead allocated through an Activity-Based Costing (ABC) system. Indirect wages were aggregated into an indirect labor overhead pool, which is allocated based on the number of shipments (“batches”) each product requires. This supports fairness in cost distribution because administrative and procurement workload scales with customer orders rather than labor hours.

Reference to Excel Workbook

All employee wage assumptions, labor classifications, annual cost calculations, and overhead allocation rates used in this section are documented in the Excel workbook. Specifically, the Assumptions and ABC Allocation worksheets contain the labor cost inputs, overhead drivers, activity-based costing rates, and supporting calculations referenced in this analysis.

Cost-Volume Profit (CVP) Analysis

Using a weighted-average selling price of approximately \$1,853 per unit and a weighted-average variable cost of \$1,323.80 per unit, the total variable cost per unit is \$1,323.80. This yields a weighted contribution margin of \$529.52 per unit, or a contribution margin ratio of about 28.57%. With annual fixed costs of \$225,193, the firm's break-even point is approximately \$788,174 in revenue, which corresponds to about 425 weighted-average units given the current product mix.

Product Mix Optimization

Purpose

The purpose of the product mix analysis was to determine how Aztec Airwalk should allocate production capacity across its product portfolio to maximize overall profitability, given its cost structure, demand assumptions, and operational constraints.

Decision Framework

The group evaluated the product mix using a contribution margin-based analysis rather than unit sales volume alone. Each product's contribution margin was calculated as the selling price minus total variable costs, including bill-of-materials costs, labor, overhead allocations, and embedded shipping costs. This approach aligns with cost-volume-profit (CVP) theory and ensures that fixed costs are covered efficiently.

Because production capacity is finite, contribution margin was analyzed on a per-unit and per-constraint basis (primarily labor and handling intensity). Products with a higher contribution margin per unit provide greater coverage of fixed costs and therefore receive priority in the recommended mix.

Resource Constraints Considered

The model does not put a hard cap on labor hours, but it factors in capacity constraints by:

- Differing labor and handling intensity across products
- Shipping weight and logistics complexity are embedded in variable costs.
- Monthly production pacing is reflected in the FP&A forecast.

These constraints are managed by prioritizing products that generate stronger margins while requiring fewer resources per unit, rather than maximizing total units produced.

Rationale for Product Focus

The analysis shows that core ramps have much higher contribution margins than accessories, despite higher shipping and material costs. Larger ramps are heavier and costlier to ship, but higher prices offset these factors and still yield strong margins.

Skatelite and coping upgrades are incremental margin enhancers, not primary drivers. They slightly raise profits without adding complexity, but do not replace core ramp production in the mix.

Recommended Product Mix

The recommended mix focuses on:

- Core ramps with strong contribution margins
- Controlled production volumes aligned with demand assumptions

- Accessories as secondary, margin-accretive offerings

This mix maximizes total margin while keeping output flexible for changing costs or demand.

Reference to Excel Workbook

The product mix decision is supported by the following workbook components:

- Pricing & Unit Economics table showing contribution margin by product
- Monthly FP&A tab summarizing contribution margin by product and period
- CVP Summary tab demonstrating the impact of product mix on break-even volume and profitability

These tables show how each product contributes to total margin and how the selected mix supports fixed costs recovery.

Facilities and Fixed Costs

Aztec Airwalk's facility-related fixed costs play a crucial role in determining the company's overall cost structure and break-even performance. These costs are largely independent of production volume and must be recovered through the contribution margin generated by the product mix.

Warehouse Rent and Utilities

The company requires a 2,000 square foot industrial workspace in the San Diego region. Market research using commercial listings on LoopNet and similar platforms indicates typical industrial rental rates of approximately \$1.50 per square foot per month, resulting in the following annual rent:

$$\text{Annual Rent} = 2,000 \text{ sq ft} \times \$1.50 \times 12 = \$36,000$$

Additional facility-related costs include:

- Electricity: \$9,000 per year
- Water: \$1,200 per year
- Insurance (property + liability): \$4,800 per year
- Internet and communication services: \$1,200 per year

Utility and insurance estimates were benchmarked using industry averages for small-scale manufacturing operations in Southern California.

Equipment and Depreciation

Aztec Airwalk requires equipment necessary for woodcutting, drilling, framing, and materials handling. Key equipment includes:

- Table saws
- Drill presses
- Air compressors
- Workbenches
- Material racks
- A forklift for pallet handling

Using straight-line depreciation with useful lives between 5 and 8 years, the total annual equipment depreciation equals \$353.

Total Facility Overhead

Combined fixed costs sum to:

$$\text{Total Facility Overhead} = 36,000 + 16,200 + 352.5 = \$52,553$$

This amount forms the facility overhead cost pool in the ABC model.

Overhead Allocation in ABC

Facility overhead is allocated based on direct labor hours, which reflect the extent to which each product consumes production space, tools, and utilities. With 6,240 available direct labor hours annually, the facility overhead rate is:

$$\text{facility overhead rate} = \frac{52553}{6240} = 8.4 \text{ per labor - hour}$$

Products requiring more labor hours absorb more facility overhead accordingly.

Role in Profitability and Break-Even

These fixed costs contribute directly to Aztec Airwalk's break-even threshold. Because fixed overhead does not vary with production volume, it must be fully covered by the contribution margin generated from product sales. Higher fixed costs increase the break-even point and require careful product mix and pricing decisions.

Reference to Excel Workbook

All facility-related cost inputs, including warehouse rent, utilities, insurance, and equipment depreciation, are documented in the Assumptions worksheet of the accompanying Excel workbook. The resulting fixed overhead amounts flow directly into the overhead rates applied in the ABC Allocation worksheet and are incorporated into the fixed-cost structure reflected in the CVP Summary sheet.

Scenario Analysis and Sensitivity

Purpose

The purpose of the scenario analysis is to evaluate how changes in key operating assumptions affect Aztec Airwalk's profitability and overall financial performance. Rather than relying on a single forecast, this section tests how sensitive results are to changes in costs and demand that are realistically expected in this business.

Decision Framework

The analysis focuses on a small set of core drivers that have the largest impact on contribution margin and break-even performance. These drivers include material costs, labor rates, shipping costs, and unit sales volume. Each driver was adjusted independently so that its impact could be observed without introducing overlapping effects from other variables.

This approach follows cost-volume-profit (CVP) logic by isolating how variable cost changes and demand shifts flow through contribution margin before affecting fixed cost recovery and profitability.

Scenarios Evaluated

Several scenarios were examined to understand downside risk and upside potential. These include increases in raw material prices, higher labor and shipping costs, and reduced demand in higher-cost shipping regions such as the Midwest and Northeast. In each case, the model recalculates unit contribution margin, total contribution, and implied break-even volume.

In contrast, an upside scenario was also evaluated to show the impact of improved demand and more favorable cost conditions. This provides a reference point for how operating leverage works in Aztec Airwalk's favor when volumes increase.

Key Findings

The results show that Aztec Airwalk's profitability is most sensitive to changes in material and shipping costs, particularly for larger and heavier ramp products. Demand changes have a meaningful effect as well, but margin erosion from cost inflation presents the greater risk to overall profitability.

Even under less favorable assumptions, the core product mix remains contribution-positive, indicating that the pricing structure provides a reasonable buffer against moderate cost increases. Favorable scenarios significantly improve margin performance and reduce the break-even volume, highlighting the importance of volume stability.

Reference to Excel Workbook

The scenario analysis is supported by the following workbook components:

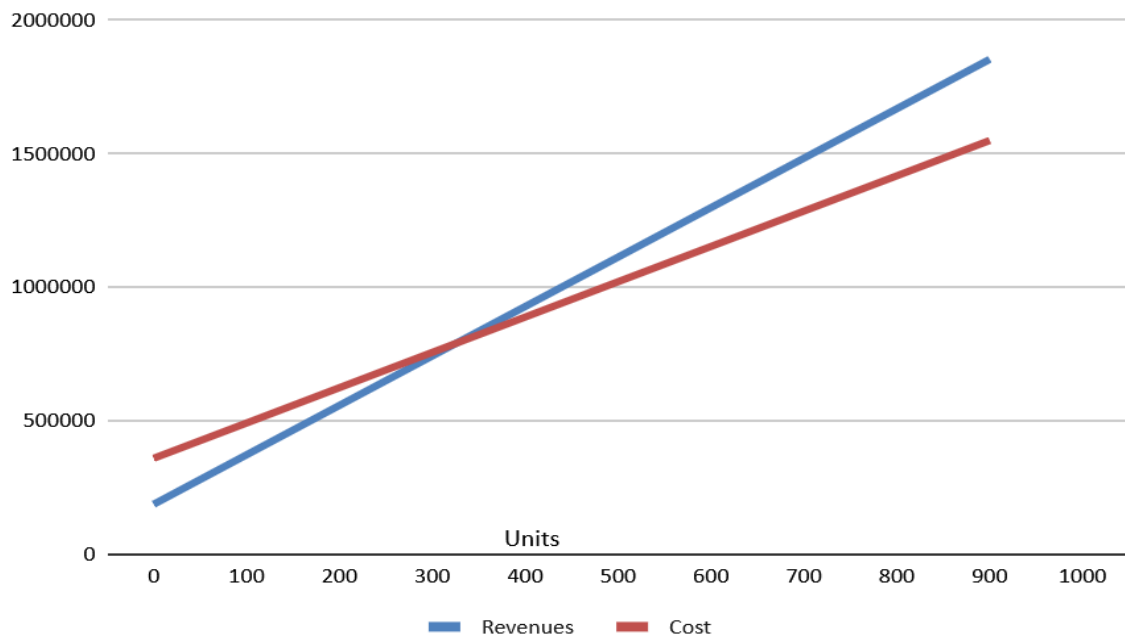
- The **Scenario Inputs** section, where cost and demand multipliers can be adjusted

- The **Monthly FP&A** tab, which shows the impact of each scenario on revenue, variable costs, and contribution margin
 - The **CVP Summary** tab, which illustrates how scenario changes affect break-even volume and fixed cost coverage
- Together, these tables show how sensitive Aztec Airwalk's financial results are to

changes in underlying assumptions and provide transparency into the drivers of profitability.

Graphs and Charts

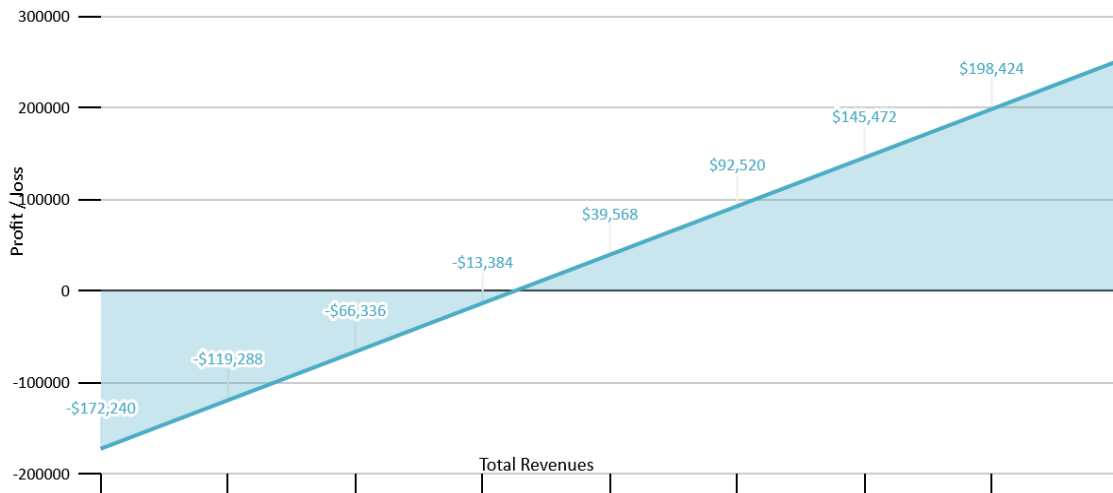
Figure 1. Break-Even Point.



The revenue - cost graph shows that Aztec Airwalk reaches its break-even point at roughly 350–375 units, where total revenues and total costs intersect. Prior to this point, costs exceed revenues due to the weight of fixed overhead. The revenue line rises at a noticeably steeper slope than the cost line, indicating a strong per-unit contribution margin consistent with the company's weighted CM ratio of 28.57%. By approximately 900 units, revenues approach \$1.67 million, while total costs reach roughly \$1.41 million, yielding an estimated operating profit of over

\$260,000 approx. This spread demonstrates the firm's favorable operating leverage: once fixed costs are absorbed, each additional unit contributes significantly to profit.

Figure 2. Profitability Graph.

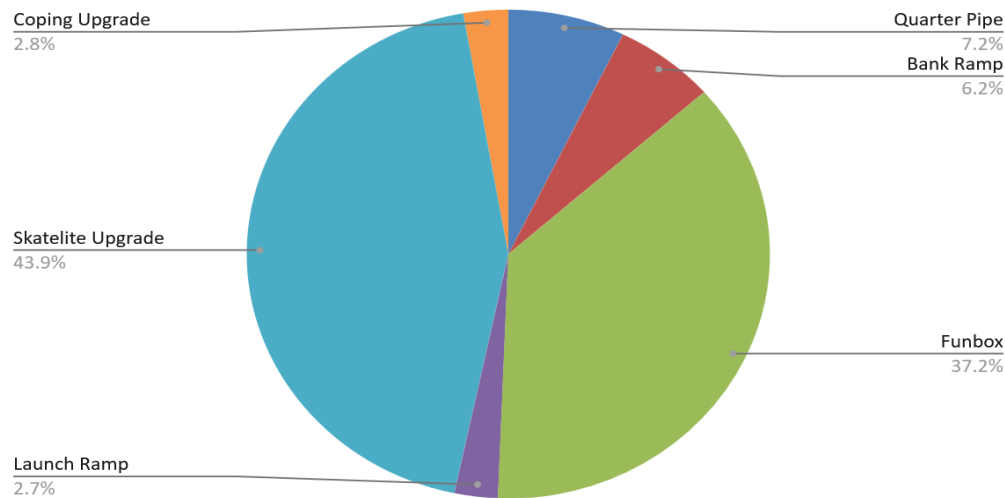


The profit/loss chart shows Aztec Airwalk's steady movement from negative to positive operating income. Losses begin at approximately –\$172,240 and narrow consistently as revenues increase, improving to –\$66,336 and then –\$13,384 just before break-even. Once profitability is reached, operating income rises predictably, reaching \$39,568, \$92,520, \$145,472, and ultimately \$198,424 at higher revenue levels. This linear progression reflects the company's stable cost structure and demonstrates how profit accelerates once fixed costs are covered.

Figure 3. Product Mix Chart.

The product mix data shows a highly concentrated contribution margin distribution. Skatelite Upgrades generate \$288,359, accounting for 38.9% of total contribution margin, while Funboxes contribute \$244,049 or 32.9%. Together, they represent over 71% of total profitability. Core ramps provide modest contributions, with the Quarter Pipe at \$47,436 (6.4%) and the Bank Ramp at \$40,991 (5.5%), while lower-margin items such as the Launch Ramp at \$17,611 (2.4%) and the Coping Upgrade at \$18,158 (2.5%) make up the smallest shares. This distribution clearly

indicates that emphasizing Skatelite upgrades and Funboxes is essential for driving overall financial performance.



Conclusion

Summary of Key Findings

Aztec Airwalk turns profitable once it sells enough units to cover its fixed costs, as shown by the cost-volume-profit (CVP) model and break-even analysis. The main products have a strong contribution margin, helping the company recover fixed costs quickly and increase operating income as sales grow. The break-even point falls within the expected sales range, which supports the idea that Aztec Airwalk's pricing and cost setup are working well.

Variable costs, especially for materials and shipping, have the biggest impact on margins. If these costs rise or fall, the contribution margin can shrink, and the break-even point can change. Keeping pricing steady and focusing on high-margin products helps protect profits. Once fixed costs are covered, every extra sale adds more to profit because of operating leverage.

Profitability and Cost Management

The core ramp products bring in the most profit for Aztec Airwalk Accessories and upgrades help increase the margin on each order, but the main ramp products are still the key to making money. Fixed costs don't change much, so managing sales volume and contribution margin is important for long-term success. Even if costs go up, Aztec Airwalk can stay profitable by keeping prices steady and focusing on the right product mix.

Recommendations to Improve Profitability

To help Aztec Airwalk become even more profitable, consider these steps:

1. **Prioritize manufacturing and selling core ramp products**, as these deliver most of the contribution margin and enable fixed-cost recovery.
2. **Manage variable costs, especially** materials and shipping, by negotiating supplier terms and optimizing packaging and logistics.
3. **Leverage accessories and upgrades to maximize** contribution per order, while avoiding capacity limits.
4. Systematically track sales volumes against break-even targets to secure operating leverage and adapt quickly to cost or demand changes.

Closing Perspective

Overall, the cost and CVP analysis back up Aztec Airwalk's pricing and product choices and show what drives its profits most. By focusing on contribution margin, keeping costs in check, and choosing the right product mix, RampCo is set up to keep making a profit as demand grows.

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